# Two laws of transfinite cardinal exponentiation

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*Editorial note.* The counted statement is the author's proposition environment labelled simplecardexpo (582-589) and the counted proof is the proof environment beginning at 591, so the pairing is by adjacency.

*Editorial note.* The counted proof is complete as the author wrote it and introduces no citation command, figure, included file or non-ASCII character, so no external reference, asset or bibliography is declared.

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit set-theory/card-arithmetic; no statement, proof step or symbol was rewritten, reordered or completed.

*Editorial note.* The statement carries two laws of cardinal arithmetic and the proof proves both, so the counted unit is the whole proposition and not one of its halves.

*Editorial scope.* One original result of the Open Logic Project unit set-theory/card-arithmetic, counted once: the two laws expressing distribution of cardinal exponentiation over cardinal addition and composition of exponents. Only the counted statement, its own author proof and the author's introductory sentence are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

*Allowed standard prerequisites (editorial).*

- Cardinal exponentiation. For cardinals a and b, the power a to the b is the cardinality of the set of functions from b to a.. Both bijections are built on the level of these function sets and establish the identities of their cardinalities (592-609).
- Cardinal addition and multiplication. The sum of two cardinals is the cardinality of their disjoint sum and the product is the cardinality of the Cartesian product of the two sets.. The first law is proved by splitting a function on the disjoint sum, and the second by indexing along a Cartesian product (593-606).
- Bijection. A bijection is a function that is injective and surjective, and two sets have the same cardinality when a bijection between them exists.. Each half of the proof ends by identifying the constructed map as a bijection between the two function sets (596-600, 607-609).

**Proposition 0.1.** *$\hat{}\mathfrak{c}a\mathfrak{c}b + \mathfrak{c}c = \hat{}\mathfrak{c}a\mathfrak{c}b \cdot \hat{}\mathfrak{c}a\mathfrak{c}c$ and $\hat{}(\hat{}\mathfrak{c}a\mathfrak{c}b)\mathfrak{c}c = \hat{}\mathfrak{c}a\mathfrak{c}b \cdot \mathfrak{c}c$, for any cardinals $\mathfrak{c}a, \mathfrak{c}b, \mathfrak{c}c$.*

*Proof.* For the first claim, consider a function $f\colon (\mathfrak{c}b \oplus \mathfrak{c}c) \to \mathfrak{c}a$. Now "split this", by defining $f_{\mathfrak{c}}b(\beta) = f(\beta, 0)$ for each $\beta \in \mathfrak{c}b$, and $f_{\mathfrak{c}}c(\gamma) = f(\gamma, 1)$ for each $\gamma \in \mathfrak{c}c$. The map $f \mapsto (f_{\mathfrak{c}b} \times f_{\mathfrak{c}}c)$ is !!abijection $\mathfrak{c}b \oplus \mathfrak{c}c\colon \mathfrak{c}a \to\!\!\to (\mathfrak{c}b\colon \mathfrak{c}a \to \times \mathfrak{c}c\colon \mathfrak{c}a \to)$.

For the second claim, consider a function $f\colon \mathfrak{c}c \to (\mathfrak{c}b\colon \mathfrak{c}a \to)$; so for each $\gamma \in \mathfrak{c}c$ we have some function $f(\gamma)\colon \mathfrak{c}b \to \mathfrak{c}a$. Now define $f^*(\beta, \gamma) = (f(\gamma))(\beta)$ for each $\langle \beta, \gamma \rangle \in \mathfrak{c}b \times \mathfrak{c}c$. The map $f \mapsto f^*$ is !!abijection $\mathfrak{c}c\colon (\mathfrak{c}b\colon \mathfrak{c}a \to) \to\!\!\to \mathfrak{c}b \cdot \mathfrak{c}c\colon \mathfrak{c}a \to$. □